

2nd Phase Problematic

Optimization of Aerial Firefighting Interventions Using Artificial Intelligence

Realized by:

Team ROBOBO

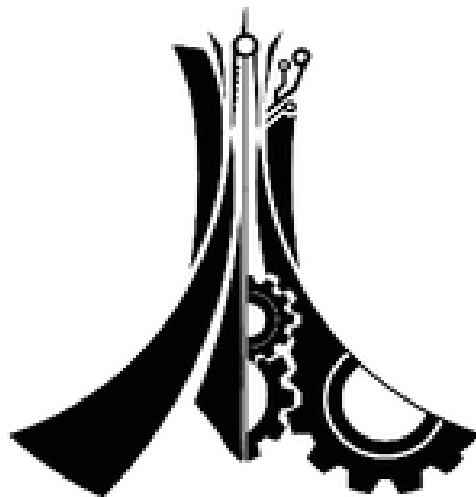


Table of Contents:

1. **General Introduction**
 - Context and Motivation
 - Applications of Autonomous Drones
 - Project Objective
2. **Chapter 1: Overview of Drones and Their Applications**
 - 1.1 Definition of a Drone
 - 1.2 Structural Components of a Drone
 - 1.3 Technological Advancements
 - 1.4 Embedded Systems and Classifications
3. **Chapter 2: System Architecture for Autonomous Wildfire Detection**
 - 2.1 Hybrid VTOL Platform
 - 2.2 Materials and Structural Integrity
 - 2.3 Environmental Resilience
 - 2.4 Regulatory Compliance
 - 2.5 Cost Estimation
4. **Chapter 3: Fire Detection Methods**
 - 3.1 Ground Sensor-Based Detection
 - 3.2 Computer Vision-Based Detection
 - 3.3 Advantages of Visual Methods
5. **Chapter 4: Real-Time Mapping with SLAM**
 - 4.1 Introduction to SLAM
 - 4.2 SLAM in GPS-Denied Environments
 - 4.3 Key SLAM Features
6. **Chapter 6: Fire and Smoke Detection Using YOLO**
 - 6.1 Overview of the YOLO Algorithm
 - 6.2 Model Training
 - 6.3 Model Integration
 - 6.4 Detection Results
7. **Chapter 7: Project Aegis – Autonomous Fire Detection and Verification System**
 - 7.1 Problem Statement
 - 7.2 Proposed Solution
 - 7.3 Technical Architecture
 - 7.4 Key Innovations

8. **Chapter 8: Project Aegis Simulation**

8.1 Objectives

8.2 Main Features

8.3 Technical Simulation Architecture

9. **Conclusion**

- Project Summary
- Limitations and Future Work
- Potential Impact

General Introduction:

In recent years, the world has witnessed the emergence of aerial robotics as one of the most active fields of embedded systems research. Autonomous aerial vehicles (UAVs) or drones have gained significant interest, a trend that continues to grow over time. This is far from coincidental; UAVs offer immense potential to assist or even replace humans in various applications. Significant technological advances in robotics, miniaturization, and electronics have led to the design and commercialization of efficient UAVs at reasonable costs, equipped with advanced capabilities that make autonomous navigation possible.

Indeed, UAVs open up numerous application possibilities for both professionals and amateurs, including:

- Mapping,
- Aerial photography,
- Video transmission,
- Military applications (reconnaissance, combat, surveillance, demining),
- Modern agriculture (gathering precise data),
- Environmental monitoring,
- Parcel delivery,
- 3D modeling of terrain and buildings,
- Wi-Fi broadcasting,
- Operations in hostile environments (radioactive zones, natural disasters, or areas inaccessible to humans),
- And recently, "UAV-Taxis" in China and Russia.

Applications in the coming years will likely expand to even more domains due to ongoing technological research worldwide.

One of the challenges in aerial robotics is designing an autonomous drone capable of "following a pre-established flight plan and avoiding obstacles." This type of drone can be employed in various civil and military applications. For instance, one could imagine a drone exploring a contaminated building or tunnel to conduct initial observations before any human intervention.

As part of this work, we will use such a drone for early wildfire detection. The drone will be configured to conduct reconnaissance missions. It will capture photos and send them to a processing center. An application implemented at the processing center will analyze the photos to detect potential fire outbreaks.

Chapter 1: Overview of Drones and Their Applications

1.1 Introduction

The development of UAVs results from scientific and technological advances in propulsion, materials, electronics, computers, sensors, navigation instruments, and more. Today, UAVs play a significant role in modern aviation. New generations of drones capable of hovering are designed to be smaller, lighter, and feature certain autonomous functions. This chapter provides a literature review, defining drones, describing their various types and navigation methods, and exploring their applications and usage regulations.

1.2 Definition of a Drone

A drone or UAV (Unmanned Aerial Vehicle) is an aircraft without a pilot on board that can fly autonomously or be remotely controlled from the ground using an onboard command system (radio control, smartphone, etc.). The term "drone" is derived from the English word for "male bee," i.e., "drone." In French, the term refers to aerial, ground, surface, or underwater vehicles, whereas the Anglo-Saxon classification distinguishes between all types of devices.

Other terms used for drones include UAS (Unmanned Aircraft System) or RPAS (Remotely Piloted Aircraft System), as used by ICAO (International Civil Aviation Organization), a United Nations body responsible for developing international air transport standards.

Drones come in various shapes and configurations depending on their mission. However, they share similar structural components, including:

1. **Motors:** Generate the energy needed for the drone's lift, considering total mass, altitude, and flight endurance. Types include turbojets, turboprops, piston engines, and electric motors.



FIGURE 1.1 – Quelques moteurs

2. **Airframes:** Maintain the drone in the air. These can be fixed (wings) or rotary (propellers) for hovering operations.



FIGURE 1.2 – Exemples de voilures.

3. **Fuselage:** Houses the motors, onboard systems, payload, and fuel or battery.



FIGURE 1.3 – Exemples d'une cellule.

4. **Onboard Systems:** Include sensors (to measure flight parameters), flight controllers, memory for flight programming, and actuators for flight control.
5. **Payload:** Equipment (sensors, processors, controllers, etc.) specific to the mission for:
 - Environmental data acquisition using mission-specific sensors (e.g., radar, optical cameras).
 - Data processing onboard for exploitation during or after the flight.
 - Transmitting useful information to the control base in real-time or storing it for later use.



FIGURE 1.4 – Exemple d'une caméra.

6. **Transmission System:** Sends data to the control base and receives commands. Communication is achieved via:
 - Relays (another aerial vehicle or satellite).
 - Optical range systems (up to 150 km) using high-power LEDs or lasers.

1.3 Technological Advancements in Drones

Embedded Intelligence (Artificial Intelligence): Advanced algorithms implemented on drones with onboard controllers offer autonomy in piloting, navigation, and mission execution. Real-time interaction with the environment makes stability and optimization critical for AI algorithms.

1.3.1 Drone System

The drone system consists of two segments:

1. **Air Segment:** Includes the drone, payload, and transmission system.
2. **Ground Segment:** Includes the control base, personnel, and equipment for launching, recovering, maintaining, and operating the drone mission.

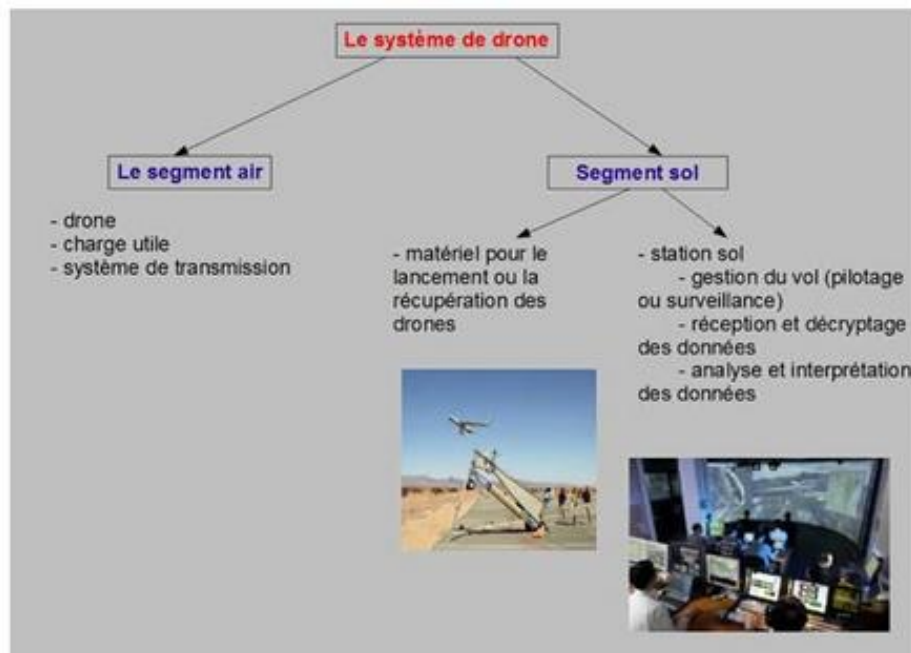


FIGURE 1.5 – Le système de drone.

Chapter 2: System Architecture for Autonomous Wildfire Detection

2.1 Platform and Configuration

The platform employs a **Vertical Take-Off and Landing (VTOL)** hybrid quadcopter design. This innovative configuration merges the superior hovering and maneuverability capabilities of a quadcopter with the extended endurance and high-speed performance of a fixed-wing aircraft. This hybrid approach allows for navigating complex terrain, confined spaces, and covering large wildfire areas efficiently.



Figure 2.1 – quadcopter drone

2.2 Materials and Structural Integrity

The chassis is constructed using heat-resistant **carbon fiber** composite materials and aerospace-grade **aluminum** components. This ensures a lightweight, robust frame that withstands extreme heat near active wildfires. Critical electronics are shielded to prevent damage from heat and embers.

2.3 Environmental Resilience and Performance

- **Ingress Protection (IP67):** Resistant to dust and temporary water immersion.
- **Wind Resistance:** Stable in wind speeds up to 60 km/h and light rain.
- **Endurance:** 90–120 minutes of flight time.
- **Speed:** Up to 120 km/h in fixed-wing mode.

2.4 Key Components and Compliance

Component	Compliant Specification	Compliance Notes (Reference to Decree 21-285)

Classification	Category 2 (2 kg < Mass ≤ 25 kg)	Based on an estimated takeoff mass of 2.5 kg – 3.5 kg.
Frame	Carbon Fiber Quadcopter Frame (450–550 mm)	Compliant material; size contributes to weight class.
Registration Marking	Physical Identity Marking Affixed to Frame	Required by Article 14: Must show the unique ID code from the National Center for Unmanned Aircraft Systems.
Flight Controller	Matek F765 or Pixhawk 6C	Required by Article 11: Must support a geovigilance system with Algerian no-fly zones loaded.
Motors / ESCs	1000–1200 KV Brushless Motors / 30A–40A ESC	Standard parts, not regulated directly, but required to be documented during homologation.
Battery	4S 5200 mAh LiPo	Standard component; impacts overall weight and classification.
Onboard Computer	Raspberry Pi 4 or NVIDIA Jetson Nano	Common professional drone component.
Cameras	FLIR Lepton 3.5 and Raspberry Pi HQ Camera	Subject to Authorization (Article 21): High-res and thermal sensors require government approval for professional usage.
Communication Module	LoRa, 4G, or Wi-Fi Module (Homologated Model)	Required by Article 32: Must be authorized for Algerian frequency use; verified during homologation.

<i>Electronic ID System</i>	Dedicated Electronic Identification Module	Required by Article 11: Must meet Algerian standards for remote ID transmission; must broadcast drone's registration number.
<i>Lighting System</i>	Luminous Signaling System (Navigation & Strobe)	Required by Article 11: Must include navigation (red/green) and anti-collision (white) lights visible from distance.

2.5 Estimated Budget and Production Costs

The following table summarizes the estimated costs associated with the complete production of a fire-detection drone, including all key hardware components, software development tools, and regulatory compliance requirements. Prices are based on mid-range, commercially available components and realistic field deployment conditions in Algeria.

A. Hardware Components

Component	Description	Estimated Cost (USD)
Frame	Carbon fiber quadcopter (450–550 mm)	\$80
Motors + ESCs	4x 1000–1200 KV brushless motors + 30A ESCs	\$80
Propellers (2–3 sets)	Spare sets included	\$20
Battery (x2)	4S 5200 mAh LiPo	\$100

Flight Controller	Pixhawk 6C or Matek F765	\$90
Onboard Computer	Jetson Nano / Raspberry Pi 4	\$120
GPS + Compass	u-blox NEO-M8N module	\$40
RGB Camera	Raspberry Pi Camera v2 / RPi HQ	\$30
Thermal Camera	Seek Thermal / FLIR Lepton / MLX90640	\$250 – \$400
RC Transmitter + Receiver	Radiomaster TX16S or similar	\$120
Telemetry Module	433 MHz telemetry kit	\$30
Additional Electronics	Power module, PDB, wiring, connectors, SD cards	\$50
Charger + Tools	LiPo charger, soldering kit, tools	\$70

Subtotal – Hardware : \$1,110 – \$1,250

B. Licensing and Legal Compliance (Algeria)

Item	Description	Estimated Cost (USD)
Drone Registration	DGAC registration fee	\$50 – \$100
Flight Authorization	Permit for surveillance / mapping use	\$100 – \$200

Optional Insurance	Public liability coverage (1 year)	\$100
Customs / Import Duties	Estimated taxes for imported parts	\$100 – \$200

Subtotal – Licensing: \$300 – \$600

Chapter 3: Fire Detection Methods

3.1 Sensor-Based Detection

To ensure early detection and rapid response to wildfires, the proposed system adopts a **multi-layered sensor architecture** that combines **ground-based environmental sensors** with **autonomous aerial investigation drones**. This hybrid approach enhances detection accuracy, minimizes false alarms, and optimizes drone deployment for verification and mapping.

A. Ground Sensor Network – Early Warning Layer

A distributed network of fixed sensors is deployed across the forest in key risk zones. These sensors operate continuously to detect abnormal environmental changes or early fire signatures. When a potential ignition is identified, the system generates an alert and transmits it to the drone base station or remote-control center.

Types of Ground Sensors Used:

- **Flame Sensors (UV/IR):** Detect the characteristic radiation emitted by flames.
- **Temperature and Humidity Sensors:** Assess the dryness and volatility of the environment, indicating fire risk.
- **Gas Sensors (e.g., CO, CO₂):** Detect combustion gases released during the early stages of fire.
- **Wind Speed Sensors:** Measure airflow conditions that affect fire propagation.
- **Soil Moisture Sensors:** Monitor ground dryness, contributing to predictive fire modeling.

These sensors are low-power, affordable, and capable of forming a mesh or point-to-point communication network using *LoRa*, *Zigbee*, or *GSM* technologies.

3.2 Computer Vision-Based Detection

Integrating computer vision into fire detection drones involves deploying advanced deep learning algorithms to process visual data from onboard cameras. This approach allows drones to :

- **Identify flames and smoke patterns** by analyzing real-time video streams.
- **Track fire spread dynamically**, providing continuous updates on fire behavior.
- **Localize fire sources precisely** within images or videos, enabling targeted interventions.

Modern AI models like **YOLO (You Only Look Once)** are particularly effective for this task. YOLO performs simultaneous detection and localization of fires in images with minimal latency, making it highly suitable for real-time missions¹. The integration of optical and infrared cameras, combined with AI, enhances detection accuracy and reduces false alarms by correlating visual data with environmental variables such as wind and topography.

3.3 Advantages of Computer Vision-Based Methods

- **Higher accuracy and scalability:** AI models can process vast amounts of data quickly, improving detection rates and allowing monitoring over large areas.
- **Visual confirmation from altitude:** Drones can identify various fire types and intensities visually, which is critical for assessing the severity and planning responses.
- **Enhanced mapping and monitoring:** Real-time visual data supports detailed fire perimeter mapping and progression tracking, aiding firefighting strategies.
- **Flexibility:** Capable of adapting to different environments and fire characteristics, including early-stage detection before flames are fully visible.

Chapter 4: Real-Time Mapping with SLAM

4.1 What is SLAM?

Simultaneous Localization and Mapping (SLAM) is a computational technique that allows a robot or drone to build a map of an unknown environment while simultaneously determining its own position within that map. This is achieved by processing data from onboard sensors such as cameras, LiDAR, or depth sensors.

4.2 How SLAM Works Perfectly Without GPS

When GPS signals are lost or unavailable—such as in dense forests, urban canyons, or indoor environments—SLAM provides an alternative navigation method by relying solely on onboard sensors and algorithms. Here is how SLAM enables reliable navigation without GPS:

- **Visual SLAM (vSLAM):** Uses cameras to capture the environment visually. Algorithms like ORB-SLAM3 extract distinctive features (keypoints) from images and track them across frames to estimate the drone's 6-DoF (degrees of freedom) pose (position and orientation) in real time.
- **Mapping and Loop Closure:** As the drone moves, SLAM builds a map incrementally. When the drone revisits previously seen locations, loop closure techniques detect this and correct accumulated drift errors, ensuring accurate localization over time.
- **Integration with Control Systems:** The SLAM output (pose and map) feeds into the drone's control architecture, enabling waypoint navigation and obstacle avoidance without external positioning systems.
- **Fault Resilience:** Advanced SLAM-based platforms incorporate fault detection and recovery mechanisms, such as rotor fault identification and emergency landing rerouting, ensuring robust operation in GPS-denied scenarios.

4.3 Key Features of SLAM for GPS-Denied Drones

- **Real-time Localization:** SLAM algorithms provide continuous pose estimation at high frequency, allowing drones to navigate dynamically changing environments.
- **Onboard Processing:** Modern implementations run on embedded systems (e.g., Raspberry Pi 4, Jetson Nano), allowing fully autonomous operation without relying on external infrastructure.
- **Sensor Fusion:** SLAM can be combined with inertial measurement units (IMUs), odometry, and other sensors to improve accuracy and robustness.
- **Map-Based Path Planning:** SLAM-generated maps enable path planning algorithms (e.g., Dijkstra's algorithm) to compute optimal routes in unknown or dynamic environments.

Chapter 6: Fire and Smoke Detection Using YOLO

6.1 Overview of YOLO Algorithm

YOLO (You Only Look Once) is a real-time object detection algorithm that has become a reference in the field of computer vision due to its speed and accuracy. Unlike traditional methods that perform region proposals and classification separately, YOLO treats object detection as a single regression problem. It divides the image into a grid and directly predicts bounding boxes and class probabilities in one pass through the neural network.

In the context of wildfire monitoring, YOLO enables fast and efficient **fire and smoke detection from aerial images**, making it well-suited for deployment onboard drones or edge computing devices such as the Jetson Nano or Raspberry Pi 4.

6.2 Dataset and Model Training

For this project, a custom YOLO model was trained using a curated dataset sourced from **Roboflow**, containing labeled images of **fire, smoke, and background scenes**. The dataset included diverse environments such as forests, urban areas, and varying light conditions to ensure robustness.

- **Dataset size:** 1,000 images
- **Classes:** Fire, Smoke
- **Model:** YOLOv8
- **Training platform:** Google Colab
- **Epochs:** 20
- **Image size:** 640×640
- **Augmentation:** Horizontal flip, scaling, color jittering
- **Optimizer:** SGD / Adam

Training was conducted using Roboflow's export tool and Ultralytics' YOLO implementation.

6.3 Model Integration

The trained YOLO model (best.pt) was exported and deployed locally for testing. It can be used in real-time scenarios by running inference on video feeds or webcam input using the following Python script:

```
[ ] from ultralytics import YOLO

model = YOLO('yolov8n.pt')
model.train(data='/content/dataset/data.yaml', epochs=20, imgsz=640)
```



```

from ultralytics import YOLO

model = YOLO('/content/runs/detect/train2/weights/best.pt')

[ ] results = model.predict(source='/content/dataset/test/images/fire-92_png.rf.35093d5b2aacdc5ba0ba65b5b7cb6146.jpg', save=True, conf=0.25)

image 1/1 /content/dataset/test/images/fire-92_png.rf.35093d5b2aacdc5ba0ba65b5b7cb6146.jpg: 640x640 6 fires, 3 smokes, 8.9ms
Speed: 2.4ms preprocess, 8.9ms inference, 1.7ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict2

```

6.4 Detection Results

After 20 epochs of training, the YOLO model demonstrated promising performance on the test dataset:

Training and Validation Losses

- **Box Loss:** Decreased from ~1.6 to ~1.3 (train) and from ~2.3 to ~2.0 (val), indicating improved localization accuracy.
- **Classification Loss:** Decreased from ~2.2 to ~1.2 (train) and from ~2.4 to ~1.8 (val), suggesting better class discrimination over time.
- **DFL (Distribution Focal Loss):** Improved gradually, showing effective bounding box regression convergence.

Performance Metrics

- **Precision (B):** Improved from ~0.42 to ~0.55, showing increasing confidence in predictions.
- **Recall (B):** Rose from ~0.40 to ~0.56, indicating the model is detecting more relevant instances.
- **mAP@0.5:** Reached approximately **0.58** by the end of training, reflecting a good balance between precision and recall at IoU=0.5.
- **mAP@0.5:0.95:** Progressed steadily to around **0.26**, which is acceptable for an early-stage fire/smoke model.

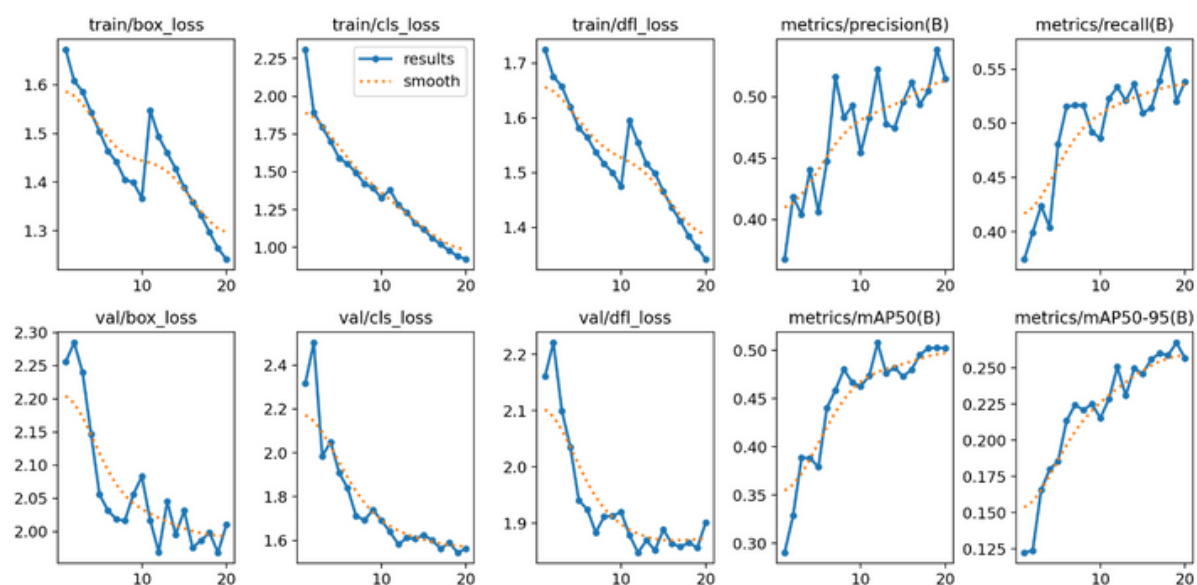


Figure 6.1: performance on the test dataset

Visual examples from the test phase showed the model correctly identifying **active flames** and **smoke plumes** in various scenarios, including low-light and occluded conditions.



Figure 6.2 – Sample detection input (fire and smoke bounding boxes)

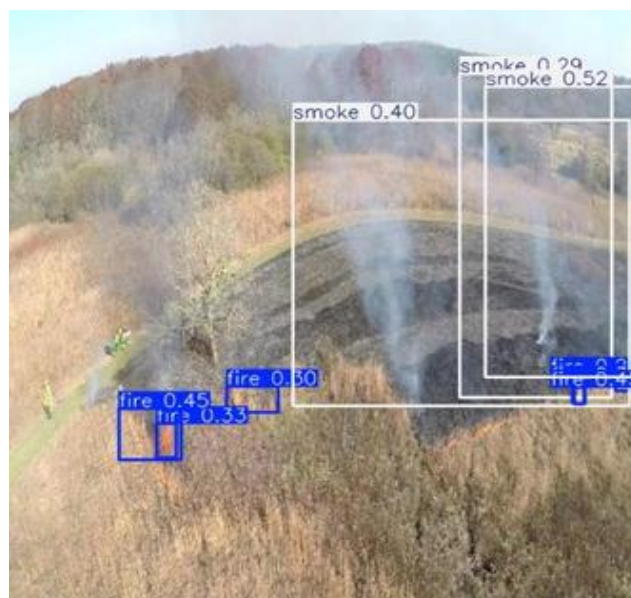


Figure 6.3 – Sample detection output (fire and smoke bounding boxes)

These results suggest that the model is **sufficiently reliable for preliminary real-world deployment**, particularly when integrated with complementary sensors such as **thermal cameras or gas detectors** for multi-modal verification. However, it is important to note that **due to time constraints**, the training process was limited to a relatively small number of epochs and dataset size. As a result, while the performance is promising, **further training and dataset expansion** would likely improve detection accuracy and reduce false positives, especially in complex or low-visibility environments.

Chapter 7 – Project Aegis: An Autonomous Wildfire Detection and Verification System

7.1 Problem Statement

Wildfires are a growing global crisis, causing billions in damages, devastating ecosystems, and endangering human lives. The single most critical factor in fire containment is **response time**: the delay between ignition, detection, verification, and intervention.

Current methods rely on outdated, manual systems such as watchtowers or public reporting, which introduce substantial delays and inaccuracy. This latency often allows small, manageable fires to escalate into uncontrollable disasters. There is an urgent need for a system capable of **detecting and verifying fires within minutes, not hours**.

7.2 Proposed Solution: Project Aegis

Project Aegis is a hybrid, two-tier system combining a network of low-power IoT sensors with a fleet of AI-powered drones to create an autonomous wildfire detection and verification infrastructure.

Two-Tier Detection Strategy:

- **Tier 1: Static "Tripwire" Sensors**
 - Deployed in high-risk zones to provide 24/7 low-power monitoring for anomalies (heat, smoke, gas).
- **Tier 2: Autonomous Drone Fleet**
 - Stationed at fixed depots (e.g., fire stations) and tasked for proactive patrols or real-time verification missions.

7.3 Technical Architecture and Methodology

A. Phase 1: Strategic Deployment

Prior to deployment, our system optimizes sensor placement and drone patrol paths based on real-world data.

1. **Risk Mapping:**
 - A multi-layer heatmap is generated using GIS data, vegetation density, historical fire records, terrain, and human activity.
 - Identifies **Critical Ignition Points (CIPs)**.
2. **Sensor Placement:**
 - A **Greedy Set-Cover Algorithm** places sensors to maximize risk coverage rather than area.

3. **Drone Routing:**
 - Identifies unmonitored high-risk areas.
 - Generates optimized patrol paths using a variant of the Traveling Salesman Problem.

B. Phase 2: Real-Time Operational Loop

The core system operates autonomously through two primary modes:

1. **Proactive Patrol:**
 - In the absence of alerts, drones autonomously patrol assigned routes.
 - Drones use thermal and optical cameras to detect potential fires and trigger Level 1 Alerts.
2. **Alert Response Workflow:**
 - A sensor detects an anomaly and sends a **Level 1 Alert** with GPS coordinates.
 - The central server calculates the **optimal drone** to respond (idle or mid-patrol).
 - A flight mission is dynamically generated and dispatched.
3. **On-Site Verification:**
 - The drone executes a pre-programmed verification routine.
 - Onboard AI confirms or rejects fire presence using high-resolution and thermal imaging.
4. **Alert Resolution:**
 - **False Positive:** Drone reports "Clear," resumes previous mission.
 - **Confirmed Fire:** System escalates to **Level 2 Alert**, notifies emergency services with:
 - GPS coordinates
 - Live video feed
 - Estimated fire size and spread
 - Access routes and nearby structures

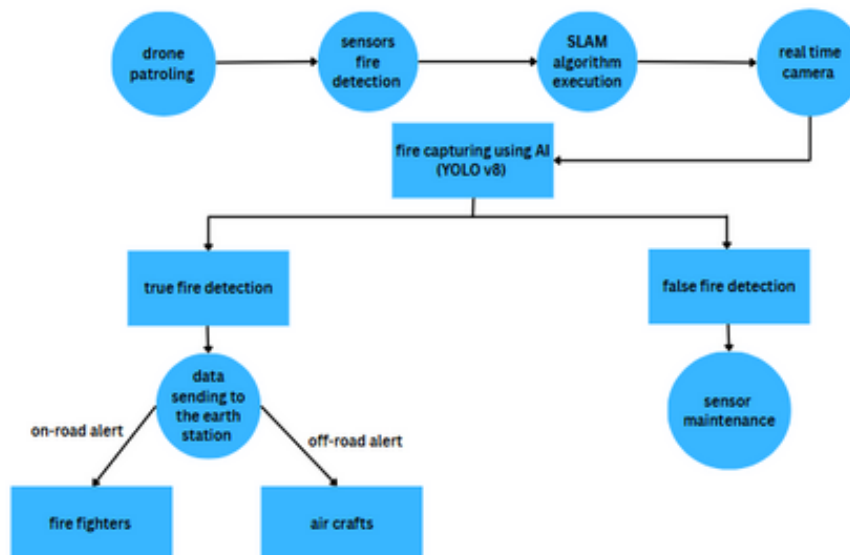


Figure 7.1

7.4 Key Innovations

- **Hybrid Intelligence:** Combines persistent ground monitoring with rapid drone verification.
- **Dual-Role Drones:** Perform both proactive patrolling and reactive response.
- **Fully Autonomous Workflow:** From detection to notification, no human input required.
- **Data-Driven Asset Deployment:** Sensor and drone placements are optimized using real risk data.
- **Actionable Insights:** Real-time intelligence empowers faster, smarter, and safer decisions.

Chapter 8 - Project Aegis Simulation

Project Aegis Simulation is an advanced, AI-driven wildfire simulation platform that integrates autonomous drones and static ground sensors to detect and respond to wildfire threats. It creates a real-time, interactive environment where users can configure forest layouts, set risk parameters, place sensors, and oversee drone deployment for wildfire detection and verification.

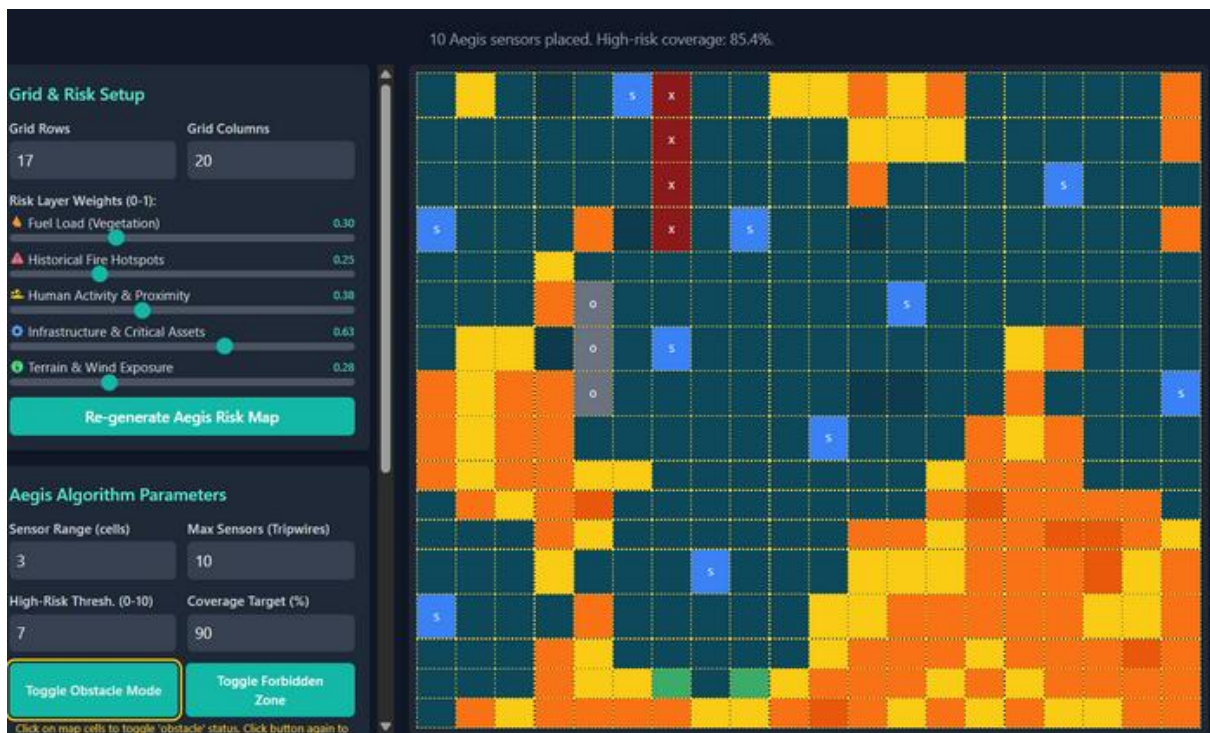
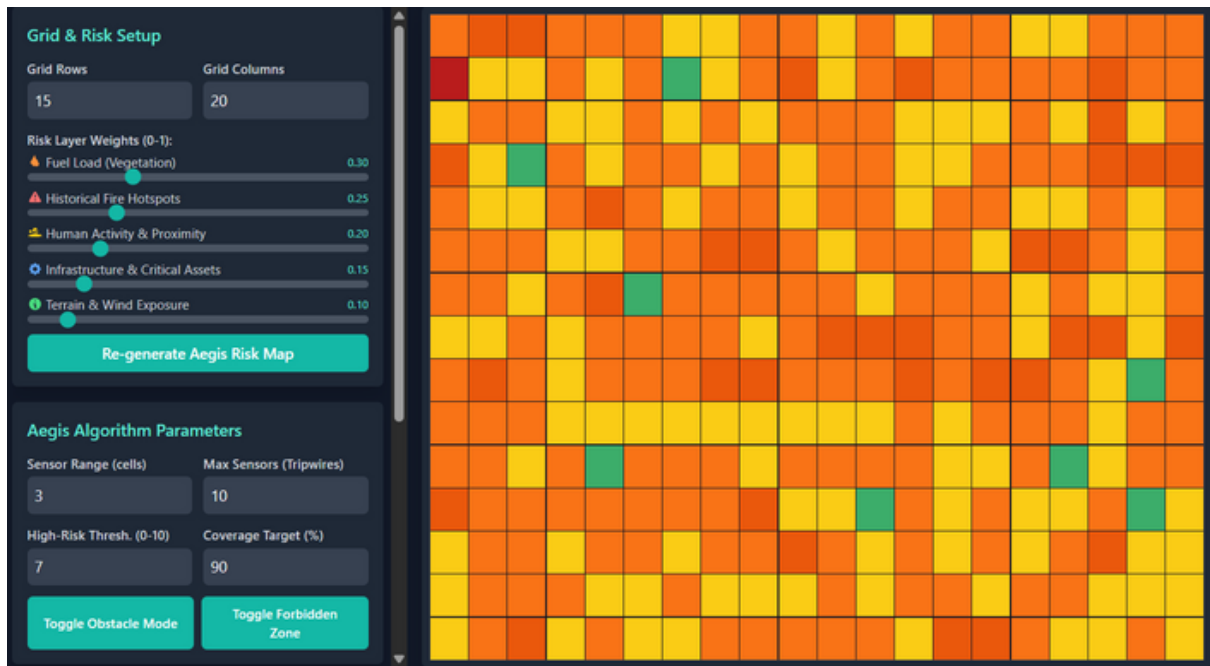
8.1 Core Objective

The goal is to simulate an efficient, AI-powered wildfire response system that reduces the time from detection to verification to under **10 minutes**, using **Gemini AI models** for intelligent decision-making, verification, and optimal drone dispatching.

8.2 Key Features

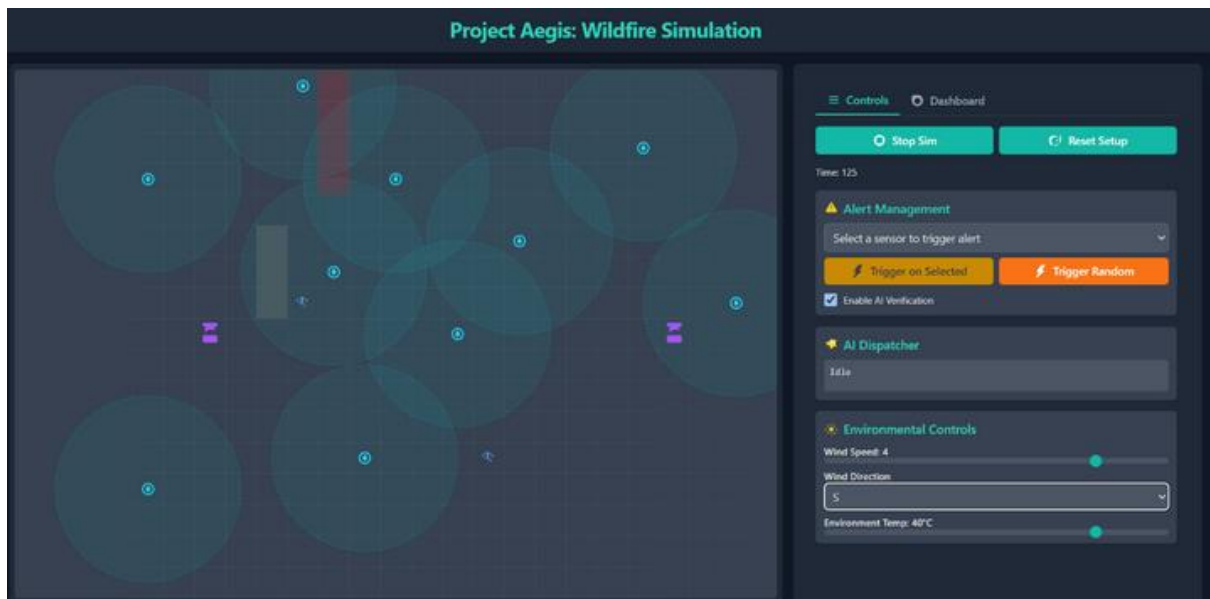
1. Forest Optimization Setup

- **Custom Grid:** Users define a grid layout of the forest (rows × columns).
- **Risk Modeling:** Adjustable weights for factors like fuel load, historical fire data, human activity, infrastructure proximity, and terrain exposure.
- **Cell Editing:** Toggle cell types (obstacles, forbidden zones, high-risk zones).
- **Sensor Placement:** Place sensors strategically; system visualizes coverage and placement effectiveness in real time.



2. Simulation Engine

- **Real-Time Alerts:** Triggered by sensors or manually by users.
- **Drone Dispatch:** AI assigns drones based on proximity, battery, and path constraints.
- **Dynamic Fire Spread:** Influenced by wind speed/direction, temperature, and fuel load.
- **Live Visualization:** See sensor zones, drone routes, fire spread, and analytics as they happen.

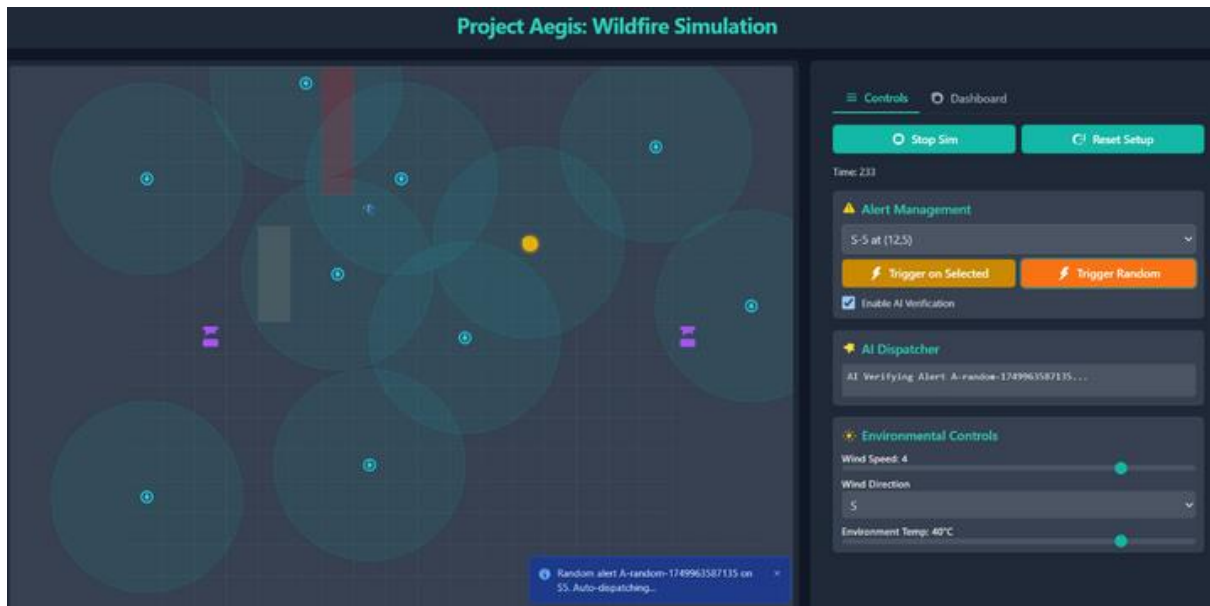


3. AI Integration

- **Gemini AI Models:** Handle drone path optimization and image-based fire verification.
- **Fire Confirmation:** AI evaluates drone-captured images with confidence scores and reasoning.

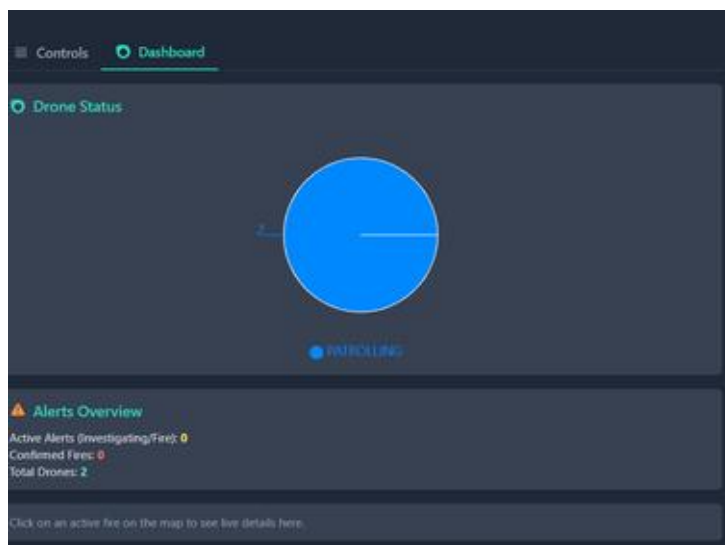
4. Control Panel

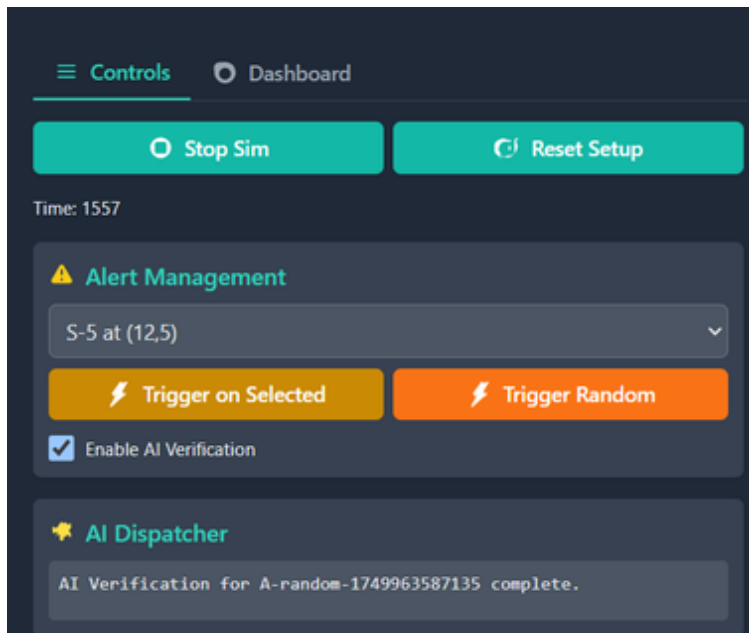
- **Simulation Controls:** Start, stop, and reset simulation.
- **Manual/Auto Alerts:** Trigger fires via sensors or user input.
- **Environment Settings:** Adjust temperature, wind speed, and direction.
- **AI Toggle:** Enable or disable AI fire verification.
- **Dispatcher Feedback:** See AI decisions and reasoning in real time.



5. Dashboards & Modals

- **Drone Dashboard:** Pie chart shows drone statuses (idle, patrolling, verifying, en route).
- **Alerts Overview:** Active alerts, confirmed fires, total drone count.
- **Fire Info Panel:** Detailed stats on selected fires (location, spread, fuel load).
- **Modals:**
 - **Verification Modal:** Drone images with AI/manual fire confirmation options.
 - **Fire Department Modal:** Simulates escalation to emergency services.
 - **Drone Info Modal:** Shows current drone details and mission status.





6. Real-Time Notifications

- Displays system messages (alerts, drone dispatches, errors) in a scrollable feed at the bottom of the map.

8.2 Technical Architecture

- **Frontend:** React (with JSX + TypeScript)
- **Styling:** Tailwind CSS
- **Bundler:** Vite (for fast build and dev experience)
- **State Management:** useReducer with AppContext for global state sharing
- **AI Integration:**
 - Text and vision models via Gemini API
 - Used for drone dispatching and fire verification
- **Simulation Logic:**
 - Fire spread based on wind, temperature, and fuel load
 - Drones manage battery, recharge at depots, and avoid forbidden zones

Conclusion

In this report, we presented **Project Aegis**, an autonomous, two-tier wildfire monitoring system that integrates a network of static IoT sensors with a fleet of AI-powered drones. This hybrid architecture was designed to overcome the critical limitations of current wildfire response systems, particularly the long delays between fire ignition, detection, and verification.

The first tier consists of **low-power environmental sensors** strategically deployed in high-risk zones, acting as constant "tripwires" for heat, gas, and smoke anomalies. The second tier, a **fleet of autonomous drones**, is responsible for both proactive patrols and rapid response missions triggered by alerts from the ground sensors.

We detailed the technical architecture of Project Aegis in two phases:

- **Strategic Deployment**, involving data-driven risk mapping, optimized sensor placement, and intelligent drone routing.
- **Real-Time Operations**, where drones autonomously perform verification missions, detect fires visually using thermal and optical imaging, and relay actionable intelligence directly to emergency services.

The system's strength lies in its ability to **reduce the detection-to-verification cycle to under 10 minutes**, enabling faster and more accurate decision-making. It also ensures that human intervention is reserved only for **confirmed threats**, thereby optimizing resource use, reducing false alarms, and enhancing firefighter safety.

While the implementation presented during this project is a proof of concept, it demonstrates the **viability and transformative potential** of combining real-time sensor data, AI, and autonomous systems to combat wildfires more efficiently. Future developments may include **hardware integration, predictive AI models**, and **drone swarm coordination** to further extend the system's scalability and reliability, especially in **GPS-denied environments**.

Ultimately, Project Aegis proposes a **proactive and intelligent shift in wildfire management**, aligning with modern challenges and offering a foundation for smarter, faster, and safer emergency responses.

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